MADIM — Model Adaptation Integrity Module

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Cognitive Reality Modeling Standard (CRMS) ·
Reality Representation Integrity Module (RRIM) ·
Model Alignment Integrity Module (MAIM) ·
Cognitive Interpretation Integrity Standard (CIIS) ·
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INTEGROS® — Integrity Standard
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Canonical Language: English (UCL)

Canonical Definition

Model Adaptation Integrity Module (MADIM) defines the structural conditions under which reality models remain capable of legitimate modification, refinement, correction, expansion, reduction, and structural adaptation in response to changing reality conditions while preserving traceability, accountability, coherence, and governance validity.

MADIM governs model adaptation.

The module ensures that cognition remains capable of updating reality models when reality changes, while preventing uncontrolled model instability, arbitrary adaptation, adaptation paralysis, or governance-invalid model evolution.

A system satisfies MADIM only if:

- adaptation remains traceable
- adaptation triggers remain identifiable
- adaptation logic remains reviewable
- model changes remain accountable

- adaptation outcomes remain assessable
- adaptation remains governance-valid

A cognition environment that cannot legitimately adapt its reality models does not satisfy MADIM.

Module Operational Role

MADIM defines the model-adaptation governance layer of CRMS.

Its role is to preserve valid reality-model evolution as operational reality changes.

Module Operational Space

MADIM governs:

- model adaptation
- model correction
- model refinement
- reality-driven updates
- adaptation traceability
- adaptation governance
- model evolution
- adaptive reality modeling

The module applies wherever cognition modifies internal reality representations over time.

Module Function

The module applies wherever systems must preserve:

- adaptive reality models
- accountable model evolution
- traceable model changes
- governance-valid updates
- operational relevance
- reality-connected adaptation

Its function is to ensure that reality models evolve in response to reality rather than remaining static or changing arbitrarily.

Minimum Implementation Framework

1. Define the Model Adaptation Object

The organization must define which reality models require adaptation governance.

This may include:

- world models
- operational models
- forecasting models
- simulation models
- strategic models
- environment models
- autonomous-agent models
- digital twins

2. Define Model Adaptation Conditions

The system must define the conditions under which adaptation remains valid.

This includes:

- adaptation triggers
- modification requirements
- validation requirements
- accountability requirements
- traceability requirements
- governance-valid adaptation conditions

3. Define Adaptation Degradation Detection Logic

The system must define how adaptation degradation is identified.

This may include:

- adaptation paralysis
- arbitrary modifications
- uncontrolled model evolution
- invalid update logic
- adaptation inconsistency
- reality-disconnected changes

4. Define Operational Response or Governance Logic

The system must define governance logic for adaptation-integrity failures.

Governance response may include:

- adaptation review
- model validation
- update restriction
- governance intervention
- escalation
- model rollback
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of:

- adaptation events
- update triggers
- model changes
- validation activities
- governance reviews
- resulting model states

A cognition environment must not remain adaptation-valid if materially significant model changes cannot be justified, reconstructed, or governed.

Use Case 1 — Autonomous Learning Agent

Scenario

An autonomous agent continuously updates its understanding of operational conditions based on newly observed events and outcomes.

Application

MADIM governs adaptation logic, update accountability, and model-evolution traceability throughout learning activities.

Result

The environment gains stronger learning reliability, improved model evolution,
and reduced risk of uncontrolled adaptation.

Use Case 2 — Enterprise Forecasting Platform

Scenario

A forecasting platform continuously updates predictive models as market conditions,
operational realities, and environmental variables evolve.

Application

MADIM governs how forecasting models adapt, what triggers modifications,
and how adaptation decisions remain accountable.

Result

The organization gains stronger forecast resilience, improved adaptability,
and reduced exposure to outdated reality models.

Canonical Closing Statement

Model Adaptation Integrity Module (MADIM) defines the structural conditions
under which reality models remain capable of legitimate modification, refinement, correction,
expansion, reduction, and structural adaptation in response to changing reality conditions
while preserving traceability, accountability, coherence, and governance validity.

Reality models cannot remain valid if they cannot adapt.
Model adaptation therefore becomes a foundational integrity condition
of governable reality modeling.

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